Operating Systems - Processes and Threads. A process is a program in execution with its own address space, open files and execution state. The operating system represents every process with a process control block that stores the program counter, the registers and the scheduling state.

A thread is the smallest unit of CPU scheduling inside a process. Threads of one process share the same address space and open files but each thread keeps its own stack and registers. Creating a thread is cheaper than creating a process because memory does not need to be copied.

CPU scheduling decides which ready process receives the processor. Round robin scheduling gives every process a fixed time quantum and rotates through the ready queue. A very small quantum causes frequent context switches and wastes processor time, while a very large quantum behaves like first come first served scheduling.